

Multiple Choice

1. **Answer:** D

Options: A) They are the shattered remains of collisions between planets.; B) They are chunks of rock or ice that condensed long after the planets and moons had formed.; C) They are chunks of rock or ice that were expelled from planets by volcanoes.; D) They are leftover planetesimals that never accreted into planets.

Note: Correct option: D - They are leftover planetesimals that never accreted into planets.

2. **Answer:** C

Options: A) 1.7 million light years; B) 2.1 million light years; C) 2.5 million light years; D) 3.2 million light years

Note: Correct option: C - 2.5 million light years

3. **Answer:** D

Options: A) stability of particles within rings.; B) gravitational effects at ring edges as the moons pass by.; C) gaps between rings.; D) Moons contribute to all of the above.

Note: Correct option: D - Moons contribute to all of the above.

4. **Answer:** D

Options: A) it is about 20000 times stronger than Earth's magnetic field; B) it traps charged particles from Io's volcanoes in a "plasma torus" around the planet; C) it causes spectacular auroral displays at Jupiter's north and south poles; D) it switches polarity every 11 years

Note: Correct option: D - it switches polarity every 11 years

5. **Answer:** B

Options: A) nuclear fusion in the core; B) by contracting changing gravitational potential energy into thermal energy; C) internal friction due to its high rotation rate; D) chemical processes

Note: Correct option: B - by contracting changing gravitational potential energy into thermal energy

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'There is no solar day Planet X is tidally locked'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'This type occurs in binary systems.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The altitude of the NCP is the same as your latitude.'

Long Form Response

11. **Answer:** Asteroids have substantial motion relative to the background stars. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** They formed as gas giants beyond the frost line and then migrated inwards.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key